OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE) III - Semester (MAIN) Examinations Dec-2024/Jan 2025

COMPUTER ORGANIZATION AND MICROPROCESSORS

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** is compulsory and answer any **four questions** from the remaining questions (2- 7), each Question carries **12** Marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any, may suitably be assumed.***

		Marks	BT	C
1. a)	Differentiate between hardwired and micro programmed control.	2	1	
b)	Mention the various types of Register Transfer Instructions.	2	2	
c)	What is Risk pipeline useful for.	2	2	
d)	Specify the sequence of micro operations that will perform the operation: i) IR← M[PC] ii) AC← AC+TR iii)DR←DR+MC	2	1	
e)	Explain the flag register of 8085.	2	2	
f)	Enlist the instruction set of 8051.	2	1	
2. a)	Explain non-restoring algorithm with an example.	6	2	
b)	Draw and explain the logic diagram of a 4-bit adder/Subtractor and its truth-table.	6	3	
3.	Explain various phases of Instruction cycle in detail with flowcharts.	12	2	
4. a)	Explain register stack and memory stack organization in detail.	6	2	
b)	Explain three address, two address, one address and zero address instructions with examples.	6	3	
5. a)	Draw the internal block diagram of 8085 with brief explanation.	6	3	
b)	Explain the timing diagram of memory read with suitable diagrams.	6	3	
6. a)	Explain the addressing mode of 8051.	6	2	
b)	Expain LCD and ADC block diagrams with suitable diagrams.	6	2	

(P.T.O.)

7. a) Explain DMA controller with example.

b) Give the differences between RISC and CISC.

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